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SUMMARY

GPU kernel engineer (M.S. ECE, UW-Madison, Dec 2025) who finds what limits **LLM inference** kernels, measures it and upstreams the result. **CUDA**, **Triton**, **Nsight**, **roofline**. 3 PRs merged in NVIDIA cuDNN and flash-linear-attention.

OPEN-SOURCE EXPERIENCE

GPU Kernel Contributor Mar 2026 - Present

- **NVIDIA cuDNN frontend, tests merged** (#1287). Prototyped native grouped-query attention (GQA) on A100 (#1223), removing K/V head copies that at 64/8 heads took longer than the kernel (7.28 vs 6.0-6.7 ms). At a 128K KV cache, 6 batched shapes ran **2.6-3.3x faster** and 10 more stopped running out of memory. NVIDIA then shipped native GQA in #1239 and merged my tests, which catch 3 injected bugs its SM80 tests missed.
- **flash-linear-attention, merged** (#1319). Continued #599 to add stick-breaking attention as **Triton** kernels. Rewrote the backward to fix a silent fp16 zero gradient caught in review. Moved to 64x32 tiles, which fit 57-89 KiB of shared memory where 64x64 needed 129-193 KiB, and reordered launches for a 1.11-1.82x faster forward. Together these reached **1.29-1.48x** the PR's previous head. **Nsight Compute** points backward cost to fp32 scans at about 12% occupancy.
- **flash-linear-attention, merged** (#1412). Added **KV-cache decoding**, tested against the uncached forward. Flagged where it slows at batch 1 and 64K keys: 32 programs on 132 SMs take **4.3 ms** while K/V reads take about 0.06 ms.
- **NVIDIA TransformerEngine, in review** (#3591). Added a **CUDA** kernel and fixed the PyTorch binding for grouped **NVFP4 dequantization**, reading offsets on device so it runs in a CUDA graph. Output matches per-tensor results bitwise.
- **vLLM, not merged** (#38338). Built a **Triton NVFP4 GEMM** reference. Ordered grouped tiles to reuse A row strips in L2 and autotuned tile sizes, holding **1,014-1,076 TFLOP/s** on RTX 5090 up to batch 49,152 where the first revision fell to 418. Benchmarked against **CUTLASS** on Llama QKV with hot L2: 1.38-3.04x up to batch 128, 0.77-0.91x above.

PROJECTS

Where the Seconds Go: MoE Decode from Host Memory (GitHub) Sep 2026 - Present

llama.cpp, C++, CUDA, Python, on RTX 5090, 4090 and 3090. Manuscript and code public.

- Built a **llama.cpp expert cache** for batch-1 MoE decode when the model doesn't fit on the GPU, with per-layer GPU slots, a decayed-frequency admission rule, GPU-signaled CPU helpers and a per-machine PCIe copy table.
- **Benchmarked it** on RTX 5090, 4090 and 3090 hosts. On RTX 5090s, gpt-oss-120b ran at **58-153 tokens/s** with 11-40% of its experts on the GPU. Across 12 configurations of 2 models, it was faster than FreeToken in 11, by up to 1.29x, and **2.0-4.0x** faster than stock llama.cpp in all 12. Swapping its admission rule for LRU ran 6-21% slower.
- Computed each machine's decode speed limit with **Belady's MIN** on real routing traces and probed read rates. None of 302 timed runs exceeded it, and at gpt-oss's smallest cache on 25 consumer machines the cache reached 31-54% of it.

Triton vs cuBLAS LLM Kernel Benchmarking (GitHub) Jan 2026 - Feb 2026

Triton, CUDA, PyTorch, on A100

- Built a tiled, autotuned **Triton** GEMM and a fused linear, bias and GELU/SiLU kernel, removing two HBM round trips and two kernel launches, and checked each against a reference before timing.
- **Benchmarked** 76 LLaMA and Mistral GEMM shapes against **cuBLAS** for latency, throughput and jitter. Bare GEMM peaked at **213 TFLOP/s**, about 68% of the fp16 tensor-core peak. In fp16, fusion raised median speed from 0.79x to 0.96x of cuBLAS and reached **1.73x** on one small-batch shape.

SKILLS

Languages and tools: C++, Python, CUDA, Triton, PTX, Metal, PyTorch, Linux, Git
GPU programming: GPU architecture, memory hierarchy, tiling, kernel fusion, Tensor Cores, CUDA graphs, HIP
Performance analysis: Nsight Systems, Nsight Compute, profiling, roofline modeling, occupancy and bandwidth analysis
LLM inference: vLLM, SGLang, cuDNN, cuBLAS, KV cache, Mixture-of-Experts (MoE), FP8, NVFP4, quantization
Hardware: NVIDIA B300, B200, H200, GH200, H100, A100, RTX 5090, Jetson Orin Nano, Apple silicon

EDUCATION

University of Wisconsin-Madison, M.S. Electrical and Computer Engineering Sep 2024 - Dec 2025

Coursework: High Performance Computing, Advanced Computer Architecture, Machine Learning, Fault-Tolerant Computing

Vellore Institute of Technology, B.Tech Electronics and Communication Engineering 2020 - 2024